ACADEMY AWARDS

DATA WAREHOUSE PROJECT

96 Ceremonies · 11,874 Nominations · 3,435 Winners · 1927 – 2023

Course	Data Warehouse
Dataset	Oscar.csv (11,876 rows, 16 columns)
Database	SQLite 3 → Star Schema
Source	Data selection, Extraction, transformation and loading
Data Visualisation	Power BI and Python
Group	G06
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Report Date	6 April 2026

Subject Description & Goals

1.1 Project Overview

This project designs, builds, and analyses a Data Warehouse (DW) centred on the complete historical record of the Academy Awards the most prestigious prize in the film industry. The source data covers every Oscar nomination from Ceremony 1 (1927/28) through Ceremony 96 (2023), encompassing 11,876 nominations, 3,435 winners, and a diverse cast of categories spanning acting, directing, writing, production, music, documentary, dance, and scientific/technical and special awards.

The project follows the complete Data Warehousing lifecycle: raw CSV ingestion into a SQLite staging table, normalisation into an operational relational model, transformation into Star Schema, analytical SQL querying, and interactive visualisation using Power BI and python

Key Dataset Facts: 96 ceremonies · 11,874 nominations · 3,435 winners · ~29% overall win rate · 1927/28 – 2023

1.2 Goals

Design a robust Star Schema suited to multi-dimensional analysis of Oscar nomination and win patterns.

Implement a complete ETL pipeline in SQLite staging, operational model, Star Schema.

Develop fact tables that incorporate both additive measures (e.g., total wins/nominations) and semi-additive snapshot measures (e.g., nominees per film category).

Enrich dimension tables with derived analytical attributes: decade and class group

Write fifteen analytical SQL queries that answer business questions about nominations, win rates, trends, top performers, and category dynamics.

Build an interactive Power BI dashboard with chart of TOP films, Winner by category, nominations by decade.

Critically reflect on the advantages and shortcomings of a DW versus an operational OLTP database

2. Planning

2.1-Dimensional Bus Matrix

Fact Table	Grain	dim_ceremony	dim_category	dim_class	dim_film	dim_ nominee
FACT_ NOMINATION	*Transactional*	✓	✓	✓	✓	✓
FACT_CEREMONY _SNAPSHOT	*Periodic Snapshot*	✓	✓	✓	—	—

2.2 Dimensions Dictionary

Dimension	Grain	Key Attributes	Analytical Enrichments
dim_ceremony	One row per ceremony (96 rows)	ceremony_key, ceremony_number, year	decade, century, era (classic / modern)
dim_category	One row per canonical category	category_key, canonical_category, category_name	Grouped by CanonicalCategory for roll-up
dim_class	One row per class (8 rows)	class_key, class_name	class_group: Creative, Technical, Other
dim_film	One row per distinct film	film_key, film_id (IMDb tt…), title	Surrogate key via ROW_NUMBER()
dim_nominee	One row per distinct person/org	nominee_key, nominee_id (IMDb nm…), name	Surrogate key via ROW_NUMBER()

2.3 Facts Dictionary

Fact Table	Grain	Additive Measures	Foreign Keys
fact_nomination	One row per nomination	nomination_count (always 1), win_count (0 or 1), multifilm flag	ceremony_key, category_key, class_key, film_key
fact_ceremony_snapshot	One row per ceremony × category × class	total_nominations, total_winners, total_films_nominated, total_distinct_nominees, avg_nominees_per_film	ceremony_key, category_key, class_key
bridge_participation	One row per nomination × nominee (many-to-many)	(associative no measures)	nom_id, nominee_key

Note:

- total_nominations, total_winners (Type: Fully Additive).
- avg_nominees_per_film (Type: Semi-Additive). Calculates the average number of people involved in a film's nomination for a specific category

3. Dimensional Data Model

3.1 Operational Model (Entity-Relationship Diagram)

Before transforming data into the Star Schema, an operational relational model was designed to normalise the raw staging data. The ER diagram below shows the six operational tables and their relationships. NOMINATION is the central entity, linked to CEREMONY (one-to-many), FILM (one-to-many), CATEGORY (one-to-many), and to NOMINEE via the PARTICIPATION bridge table (many-to-many).

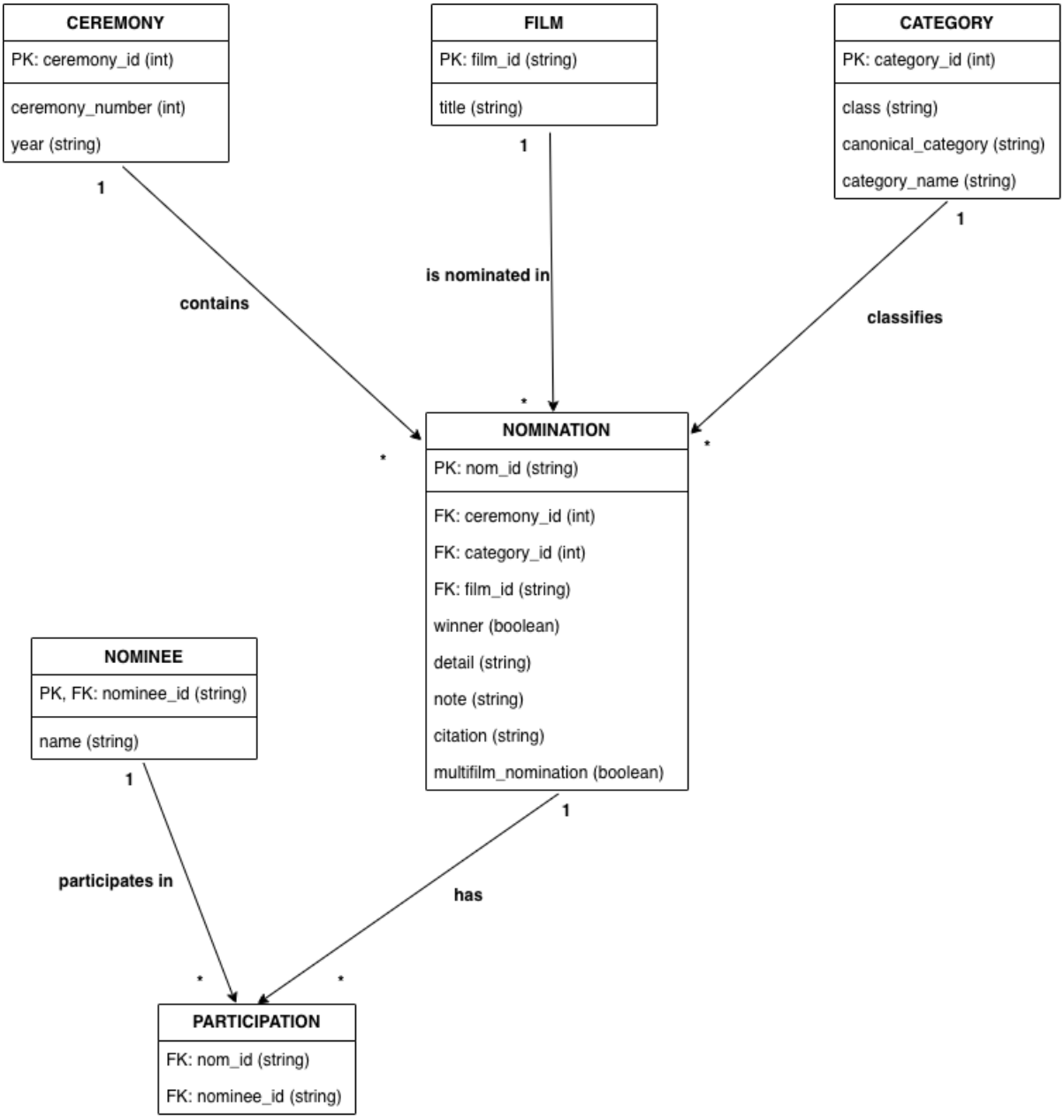

Figure 1 Operational Entity-Relationship Diagram

3.2 Star Schema (Dimensional Model)

The operational model was then transformed into a Kimball-style Star Schema for analytical use. The diagram below shows the full schema: two fact tables (FACT_NOMINATION and FACT_CEREMONY_SNAPSHOT), five dimension tables (DIM_CEREMONY, DIM_CATEGORY, DIM_CLASS, DIM_FILM, DIM_NOMINEE), and the BRIDGE_PARTICIPATION bridge table resolving the many-to-many relationship between nominations and nominees.

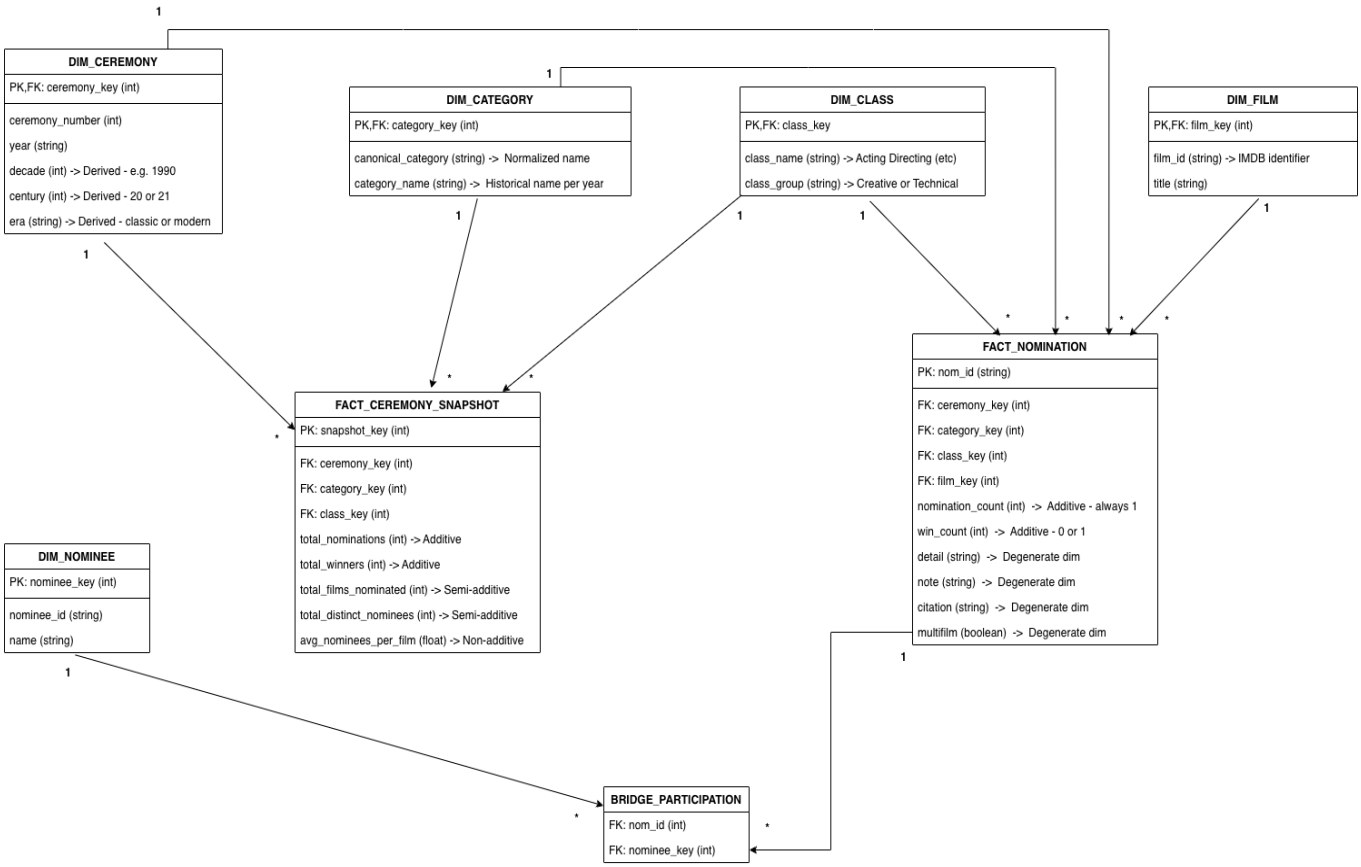

Figure 2 Star Schema Dimensional Model

3.3 Key Design Decisions

- Bridge Table for many-to-many: A single nomination can involve multiple nominees (e.g., a jointly-nominated film crew). The bridge_participation table resolves this cleanly without row duplication in the fact table.
- Snapshot Fact Table: fact_ceremony_snapshot pre-aggregates totals per ceremony/category/class combination. This reduces Power BI query time from full-table scans to a few hundred rows.
- Conformed Dimensions: dim_ceremony, dim_category, and dim_class are shared across fact_nomination and fact_ceremony_snapshot, enabling consistent slicing across both fact tables.
- Derived Dimension Attributes: decade, century, and era are computed during ETL using SQL substring and integer division adding analytical value with no change to source data.
- Degenerate Dimension: nom_id is kept as the fact primary key rather than a separate dimension table, as it carries no descriptive attributes beyond those already in other dimensions.
- SciTech as Separate Class: Scientific and Technical Awards are structurally different (non-competitive, committee-selected). Separating them into their own class group prevents them from distorting competitive win-rate analysis.

4. Data Sources, Extraction, Transformation & Loading

4.1 Data Source

The source file is oscars_8.csv a comprehensive CSV record of all Academy Award nominations from 1927/28 through 2023.

Attribute	Value
File format	CSV, UTF-8, comma-delimited
Total rows	11,876 nomination records across 96 ceremonies
Columns	16: Ceremony, Year, Class, CanonicalCategory, Category, NomId, Film, FilmId, Name, Nominees, NomineeIds, Winner, Detail, Note, Citation, MultifilmNomination
Winner encoding	1 = winner , null or 0 = not winner
Film identifier	FilmId (e.g. tt0019217, ef0050022)
Nominee identifier	IMDb-style NomineeIds (e.g. nm0001932, co0074827); comma-separated for joint nominees
Date range	Ceremony 1 (1927/28) through Ceremony 96 (2023)

4.2 Extraction (01_load.sql)

The CSV is loaded into SQLite as a staging table (stg_oscars) using a single CREATE TABLE and bulk INSERT statement. No external ETL tool is used the entire pipeline runs natively in SQLite, making it fully self-contained and reproducible. The staging table preserves all 16 raw columns with minimal typing; winner is stored as REAL to accommodate the null/1.0 encoding. No rows are filtered at this stage, ensuring complete data lineage from source to warehouse.

4.3 Transformation (02_create_load_tables.sql)

All transformations are applied inline within INSERT...SELECT statements. Key transformations include:

- Winner Normalisation: The raw winner field contains null, 0, 1, 1.0, '1', 'TRUE', 'true' variants. A CASE expression normalises all to integer 0 or 1, preventing incorrect aggregations.
- Year Parsing: Year values such as '1927/28' are parsed using SUBSTR(year, 1, 4) to extract the four-digit start year for all date-derived attributes.
- Decade Derivation: (CAST(SUBSTR(year,1,4) AS INTEGER) / 10) * 10 produces clean decade integers (1920, 1930, …, 2020).
- Era Classification: Years before 2000 are labelled 'classic'; 2000 onwards are labelled 'modern'. This binary split enables era-level comparative analysis.
- Class Grouping: Acting, Directing, Writing, Title, and Special are grouped as 'Creative'; Production and SciTech as 'Technical'. This classification underpins Q12 analysis.
- Deduplication: INSERT OR IGNORE and SELECT DISTINCT prevent duplicate dimension rows from multi-nominee nominations that share the same NomId.
- Surrogate Keys: ROW_NUMBER() OVER (ORDER BY ...) generates integer surrogate keys for dim_film and dim_nominee, decoupling the warehouse from IMDb identifiers.
- Multifilm Flag: MultifilmNomination is normalised from its mixed-type source (null, 0, 1, '1', 'TRUE') to a clean 0/1 integer.

4.4 Loading Order

Loading follows strict foreign-key dependency order (PRAGMA foreign_keys = ON):

1. stg_oscars raw staging data from CSV
2. Operational tables: ceremony, film, category, nominee, nomination, participation
3. Dimension tables: dim_ceremony, dim_category, dim_class, dim_film, dim_nominee
4. Central fact: fact_nomination (surrogate keys resolved via JOIN to dimensions)
5. Bridge: bridge_participation
6. Snapshot fact: fact_ceremony_snapshot (aggregated via GROUP BY from fact_nomination + bridge)
7. Performance indexes on all foreign key columns

5. Querying & Data Analysis

5.1 Overview

Fifteen analytical SQL queries were written against the Star Schema (03_teste_queries.sql). The queries address the five research questions from Section 1 and directly feed the Power BI dashboard.

File	Description
01_load.sql	Load staging data from oscars.csv
02_create_load_tables.sql	Operational schema + ETL + Dimensions, facts, snapshots
03_teste_queries.sql	Analytical queries
oscars.csv	Source dataset (~10k rows)

> *Dataset totals: 11,874 nominations · 3,435 winners · ~29% overall win rate across 96 ceremonies (1927–2023)*

5.2 Query Analysis

The dataset covers **96 Oscar ceremonies** from 1927/28 to 2023, with a total of **11,874 nominations** and **3,435 winners**.

5.3 Query Results

Q1 & Q2 Overall Scale

The dataset contains 11,874 nominations across all ceremonies, of which 3,435 resulted in a win. That's roughly a 29% overall win rate which makes sense given that most categories have around 5 nominees per year.

Q3 Most Nominated Films

Film	Nominations	Notes
A Star Is Born	26	Accumulated across four versions (1937, 1954, 1976, 2018)
West Side Story	18	Includes original 1961 and 2021 remake
Titanic	16	Single release exceptional single-film count
Mutiny on the Bounty	15	15 nominations, only 1 win (6.7% rate)
Moulin Rouge	15	—
Gone with the Wind	15	10 wins 66.7% conversion rate
Little Women	14	—
La La Land	14	—
Cleopatra	14	—
All about Eve	14	—

A Star Is Born tops the all-time list with 26 nominations though this is spread across four separate productions spanning 80 years. Titanic (1997) is the standout single-film achievement with 16 nominations and 12 wins (75% win rate). Gone with the Wind's 66.7% conversion rate and Mutiny on the Bounty's 6.7% rate represent opposite extremes in nomination efficiency.

Q4 Most Nominated Categories

Best Picture leads as expected it's the most prestigious and has existed since the very first ceremony. Music categories rank surprisingly high, reflecting how many films compete each year for score and song.

Category	Nominations
BEST PICTURE	601
MUSIC (Original Score)	519
MUSIC (Original Song)	475
DIRECTING	471
FILM EDITING	450
ACTRESS IN A SUPPORTING ROLE	440
ACTOR IN A SUPPORTING ROLE	440
WRITING (Original Screenplay)	415
SHORT FILM (Animated)	393
DOCUMENTARY (Short Subject)	387

Q5 Wins by Year (Top 10)

1998 is the standout year Titanic alone won 11 Oscars that night, but the high total also reflects the Academy expanding the number of categories over the decades. The 1990s and 2000s consistently show high win totals compared to earlier decades.

Year	Wins
1998	62
1994	53
2001	52

Year	Wins
1995	51
2014	50

Q6 Distinct Nominees by Class

Production and Title classes involve the most unique people Production covers cinematography, editing, art direction, and costumes, all of which involve large crews. Directing is the most exclusive class, with just 278 distinct nominees across nearly 100 years

Class	Distinct Nominees
Production	2123
Title	2049
Acting	980
SciTech	872
Writing	829
Music	650
Special	341
Directing	278

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Q8 Most Nominated Films with Win Rate

Film	Nominations	Wins	Win Rate
A Star Is Born	26	4	15.4%

Film	Nominations	Wins	Win Rate
West Side Story	18	11	61.1%
Titanic	16	12	75.0%
Mutiny on the Bounty	15	1	6.7%
Gone with the Wind	15	10	66.7%

> *Titanic's 75% win rate is one of the most efficient award-night performances in Oscar history. Mutiny on the Bounty's 6.7% from 15 nominations is among the worst conversion rates for a heavily-nominated film.*

Q9 Nominations by Decade

Decade	Nominations	Wins	Win Rate	Notes
1920s	133	33	24.8%	Only 1–2 ceremonies; limited categories
1930s	866	227	26.2%	Category expansion begins
1940s	1,528	343	22.5%	WWII documentary surge; peak single decade
1950s	1,259	355	28.2%	Studio system decline; new categories
1960s	1,235	328	26.6%	Foreign language & documentary growth
1970s	1,152	345	29.9%	New Hollywood era

Decade	Nominations	Wins	Win Rate	Notes
1980s	1,182	381	32.2%	VFX and SciTech awards expand
1990s	1,320	460	34.8%	Highest win rate decade
2000s	1,288	411	31.9%	Streaming era begins
2010s	1,394	421	30.2%	Category cap standardisation
2020s	517	131	25.3%	Incomplete only 2020–2023

The 1940s saw the sharpest rise in nominations the Academy expanded documentary categories dramatically during WWII, and the wartime boom in patriotic filmmaking flooded the nomination pool. The 1990s achieved the highest win rate (34.8%), likely reflecting more standardised 5-nominee shortlists. The 2020s figure is incomplete, covering only four ceremonies.

Q11 Top Categories by Win Rate

Category	Nominations	Wins	Win Rate
BEST PICTURE	601	96	16.0%
MUSIC	519	85	16.4%
DIRECTING	471	95	20.2%
FILM EDITING	450	90	20.0%

Most major creative categories cluster around 20% win rate, consistent with a five-nominee shortlist. Best Picture is slightly lower at 16% because the Academy expanded to up to 10 nominees from 2010 onwards, increasing competition per award.

Q12 Creative vs Technical Classes

Class Group	Class	Nominations	Wins	Win Rate
Technical	Production	3,222	683	21.2%
Creative	Title	2,706	572	21.1%
Creative	Acting	1,835	373	20.3%
Technical	SciTech	890	890	100.0%
Creative	Special	376	370	98.4%

The 100% SciTech win rate is not a data error. Scientific and Technical Awards are honorary every selected recipient wins by definition. The same applies to Special Awards (98.4%). These are not competitive nominations in the traditional sense.

Q13 Most Winning Nominees

Name	Nominations	Wins	Win Rate
Walt Disney, Producer	62	25	40.3%
Eastman Kodak Company	15	15	100.0%
Metro-Goldwyn-Mayer	68	13	19.1%
Alfred Newman	34	7	20.6%

Walt Disney leads all individuals with 25 wins, accumulated primarily through short animated films and features during the 1930s–1950s. Eastman Kodak's 100% rate reflects multiple Scientific and Technical Awards, all of which are non-competitive. MGM's 13 wins from 68 nominations (19.1%) represents decades of studio-era dominance in the nomination pool.

Q14 Top Films by Class

Class	Film	Nominations	Wins
Acting	A Star Is Born	7	0
Acting	On the Waterfront	5	2
Acting	Network	5	3

A Star Is Born accumulated 7 acting nominations across its various versions without a single acting win a remarkable record of near-misses. Network (1976) stands out with 3 wins from 5 acting nominations, one of the strongest acting sweeps in a single ceremony.

Q15 Classic vs Modern Era

Era	Nominations	Wins	Distinct Films	Distinct Nominees	Win Rate
Classic (pre-2000)	8,677	2,472	4,691	5,489	28.5%
Modern (2000+)	3,199	963	1,713	2,562	30.1%

The classic era dominates in raw volume spanning over 70 years versus 23 for the modern era. The modern era's slightly higher win rate (30.1% vs 28.5%) likely reflects more standardised 5-nominee shortlists in recent decades. Strikingly, the classic era shows far greater nominee diversity 5,489 distinct nominees versus 2,562 modern, even accounting for the era's larger size. This suggests the modern era sees a higher concentration of nominations among fewer, repeatedly recognised individuals and films.

6. Visualisation

6.1 Dashboard Overview

The Power BI dashboard was built on the Oscars data table (imported from oscars_8.csv) and presents charts, three KPI cards, and across two report pages. The visualisations were chosen to complement each other bars for ranking, a line for time trends, a grouped column for decade comparison, and a donut for proportional composition.

6.2 Chart Specifications

Chart 1 Top Films by Winner Count

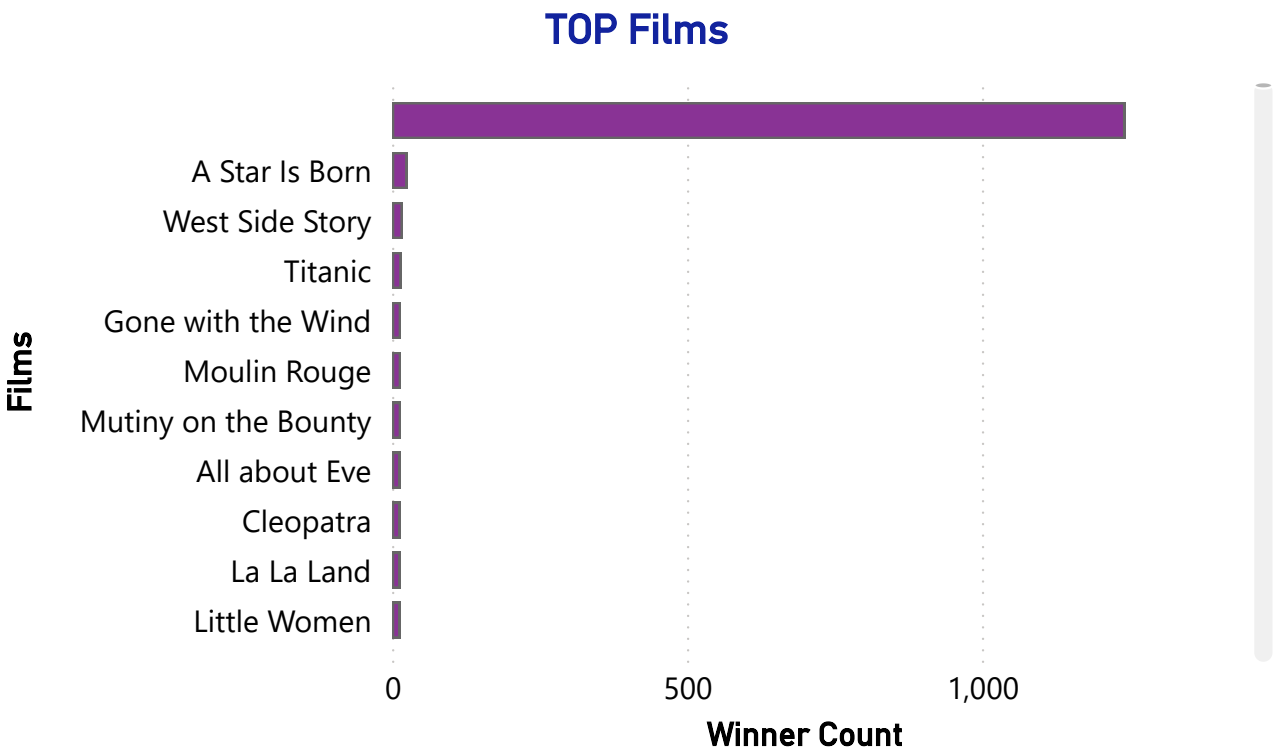


Chart 2 Academy Award Wins Over Time

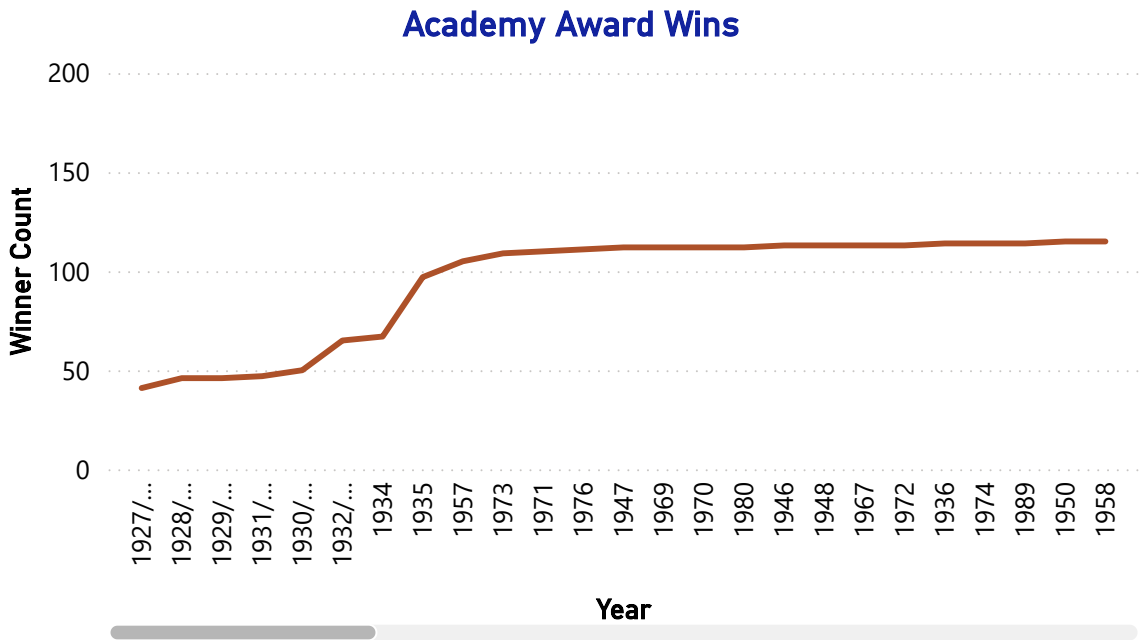


Chart 3 Top Categories by Nomination Count

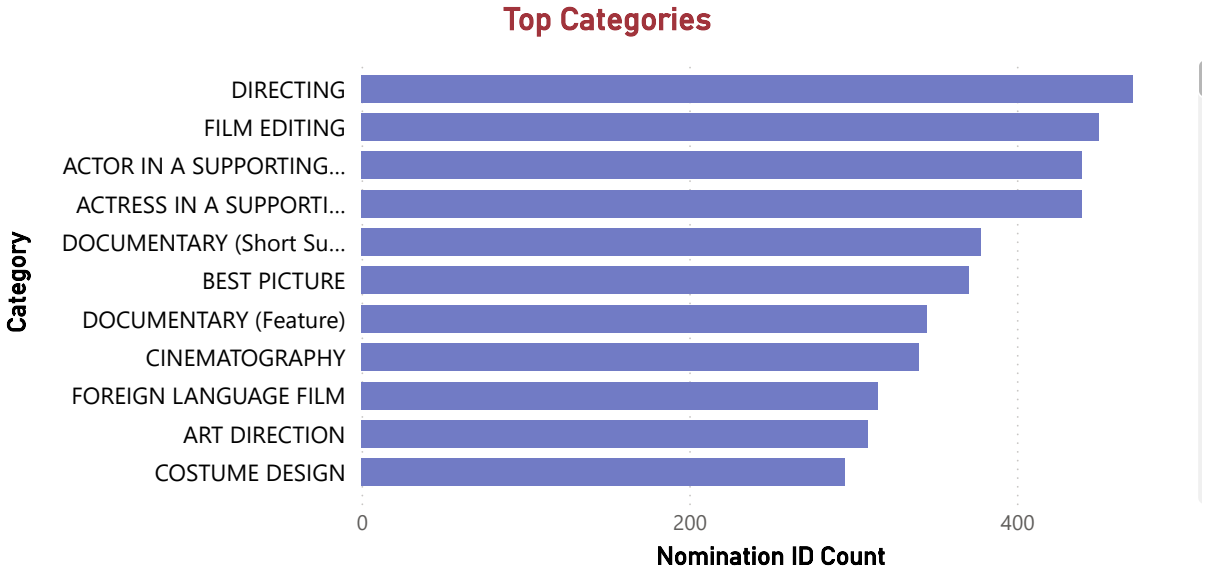


Chart 4 Win Rate by Category

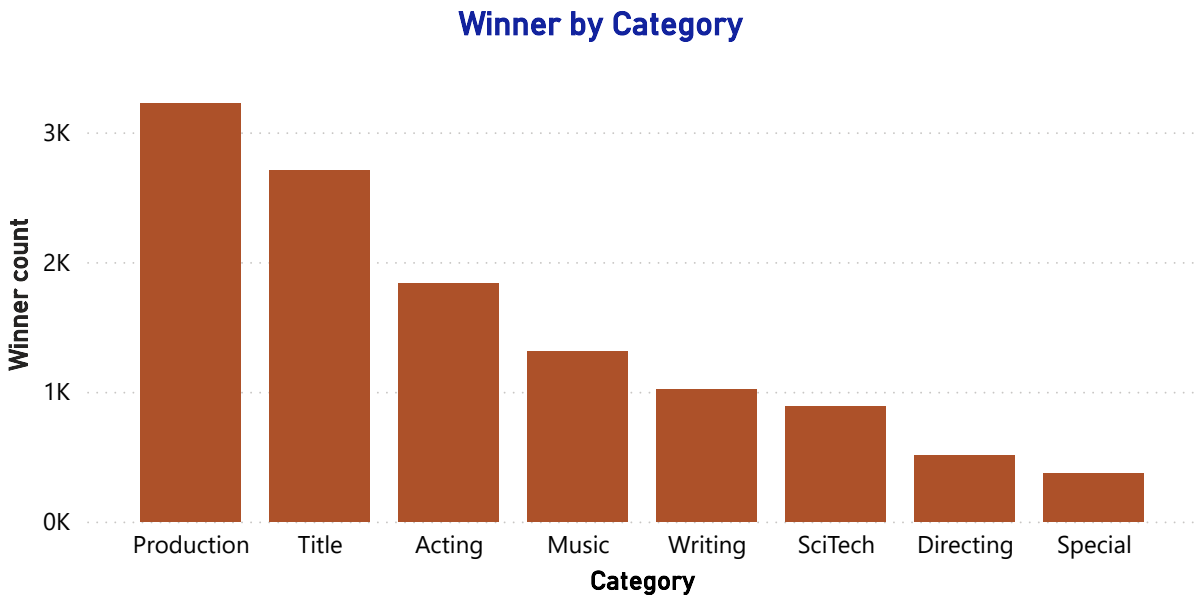


Chart 5 Nominations by Decade

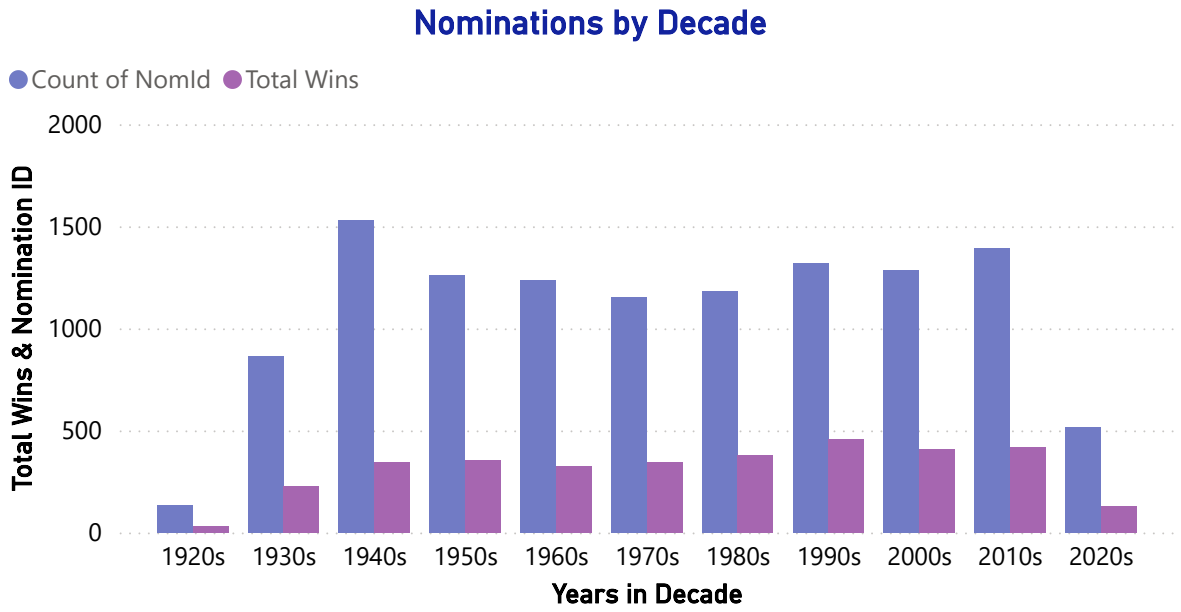


Chart 6 Winners by Category + Donut

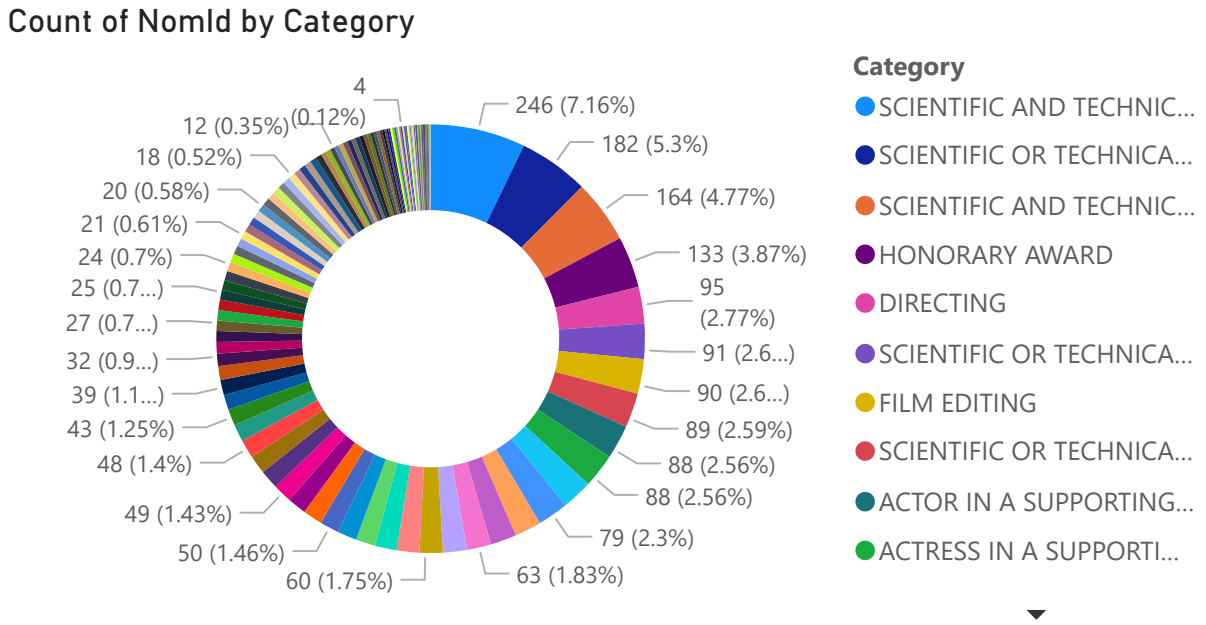
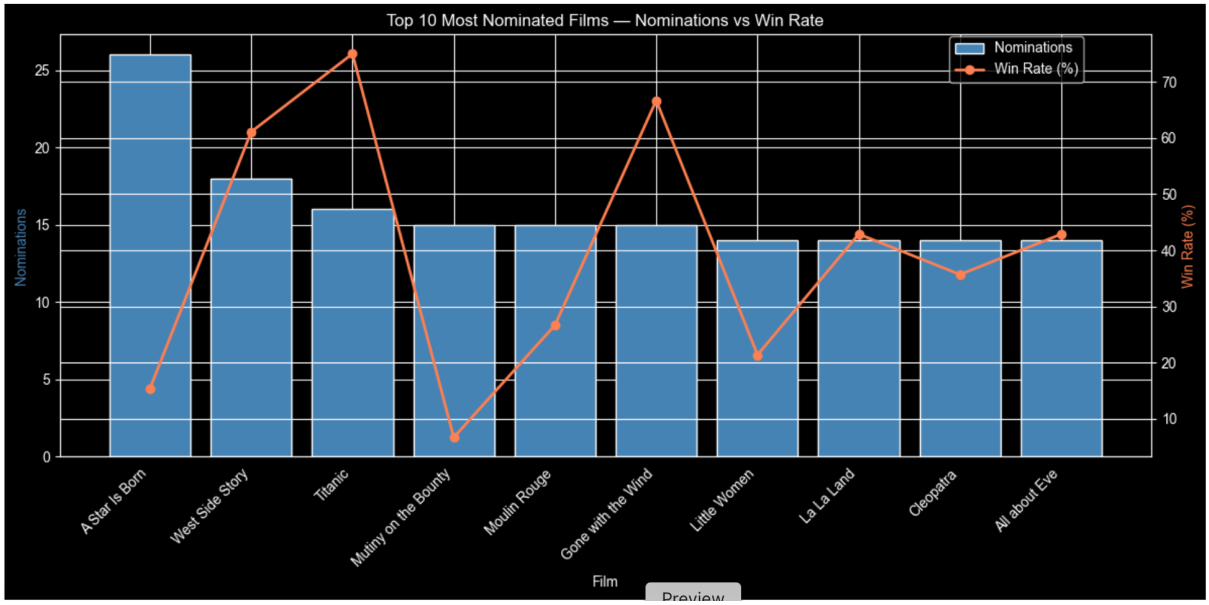


Chart 7- Nomination Vs Win Rate



7. Conclusion & Critical Reflection

7.1 Summary of Findings

This project delivered a complete Data Warehouse solution for the Academy Awards dataset from raw CSV ingestion through a normalised operational schema to a Kimball Star Schema, with analytical SQL queries and an interactive Power BI dashboard.

The five research questions from Section 1 were fully answered: Titanic holds the best single-film win rate (75%); nomination volumes peaked in the 1940s due to the WWII documentary surge and again in the 1990s–2010s due to category expansion; SciTech and Special categories distort win-rate analysis because they are non-competitive; the modern era shows a slightly higher win rate (30.1% vs 28.5%) but far less nominee diversity; and Walt Disney remains the most decorated individual with 25 wins from 62 nominations.

7.2 Advantages of a Data Warehouse vs Operational Database

Advantage	Explanation
Query Performance	Pre-aggregated fact_ceremony_snapshot returns summary statistics instantly. The equivalent OLTP query must JOIN six tables and scan 11,874 rows.
Analytical Richness	Derived attributes (decade, era, class_group) are computed once at load time and reused across all queries no repeated CASE expressions in every SQL statement.
Conformed Dimensions	dim_ceremony and dim_category can be joined to both fact_nomination and fact_ceremony_snapshot, producing consistent results regardless of which fact is queried.
Separation of Concerns	The operational schema handles data integrity and normalisation; the Star Schema handles analytics. Changes to one do not disrupt the other.
Power BI Integration	Star Schemas are the native model for Power BI's relationship engine. Conformed dimensions enable automatic cross-filtering between all visuals without manual DAX workarounds.
Historical Preservation	The warehouse stores a historical snapshot. Even if source data is corrected, the analytical record remains stable and auditable.

7.3 Shortcomings & Limitations

Shortcoming	Explanation
No Incremental Load	The ETL performs a full reload from the CSV. Adding Ceremony 97 would require re-running the entire pipeline rather than appending new rows only.
No Slowly Changing Dimensions	dim_film has no history. If Titanic's title record is corrected in the source, the old film_key loses its link to historical nominations (SCD Type 1 overwrite issue).
Single Data Source	The warehouse draws from one CSV. A production system would integrate box office revenue, critic scores, streaming data, and IMDb ratings for richer multi-source analysis.
Winner Encoding Ambiguity	The raw winner field contains null, 0, 1, and 1.0. MultifilmNomination rows and joint-nominee records required careful CASE logic; edge cases may still exist.
SQLite Limitations	SQLite lacks native PIVOT, parallel query execution, and concurrent write access. It is suitable for a single-user analytical prototype but not for production-scale deployments.
Flat Power BI Model	The Power BI dashboard was built on the flat oscarsdata staging table rather than the Star Schema. This forgoes the relational benefits of conformed dimensions and makes complex cross-table DAX measures more verbose.
2020s Incompleteness	The dataset contains only four modern ceremonies (2020–2023). Decade-level comparisons for the 2020s are therefore not statistically comparable to earlier complete decades.

7.4 Future Work

- Connect Power BI directly to the SQLite Star Schema via ODBC, enabling full use of conformed dimensions, bridge tables, and the snapshot fact eliminating the need for DAX workarounds on the flat table.
- Implement SCD Type 2 for dim_film and dim_nominee so that title corrections and name changes are tracked historically without breaking fact-table links.
- Enrich dim_film with external attributes box office gross, genre, runtime, Rotten Tomatoes score via IMDb FilmId keys to enable correlation analysis (e.g., does commercial success predict more nominations?).
- Add incremental ETL logic to append each new ceremony's data without full reload, enabling the warehouse to be updated annually in minutes.
- Extend the Power BI dashboard with Python-based predictive visuals to forecast nomination volumes and win probabilities for future ceremonies.